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CS260A: User Interface Design & Development
Professor Bjoern Hartmann
2 October 2025

Team Brainstorm Writeup

Team Name

Open-Minded Oski

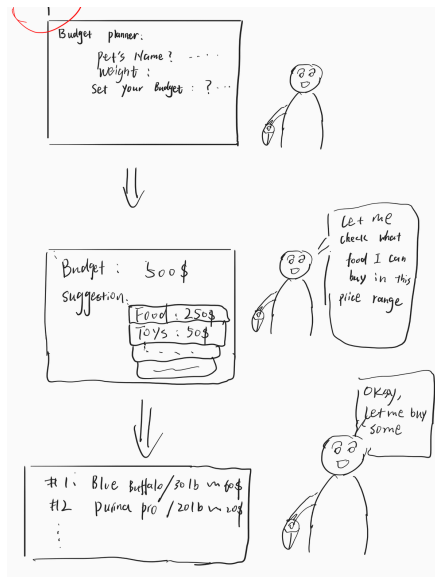
Each Team Member's Name

Shanaya Malik, Minghui Wang, Carrie Wan

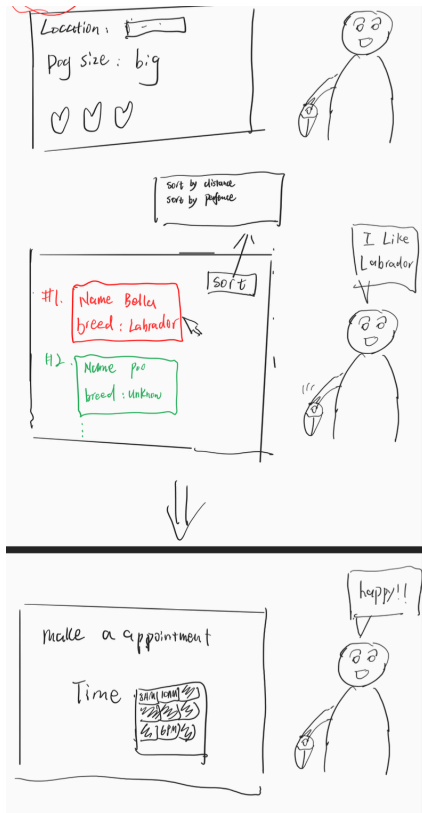
Brainstorm

User Group: Pet Owners

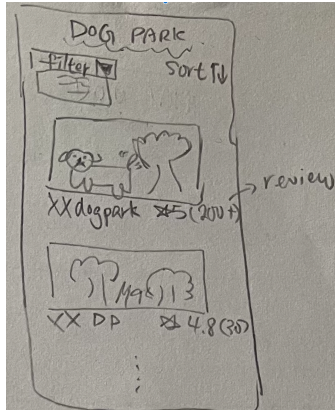
1. An AI app could teach new pet owners how to care for their pets by analyzing photos and providing tailored advice.
2. A pet health monitor could detect changes in posture, fur, or eating habits that might indicate illness.
3. A budgeting and planning tool could predict annual pet expenses and suggest cost-saving strategies.



4. An AI-powered training coach could analyze owner voice commands and recommend adjustments for better obedience.
5. A behavior translator could interpret pet signals like barking or tail movement into likely needs or emotions.
6. A pet camera assistant could analyze pets' emotions and suggest comfort strategies when owners are away.
7. A shelter connector app could match prospective adopters with local rescues and shelters.



8. A park-finding assistant could recommend safe, nearby dog parks based on location and reviews.



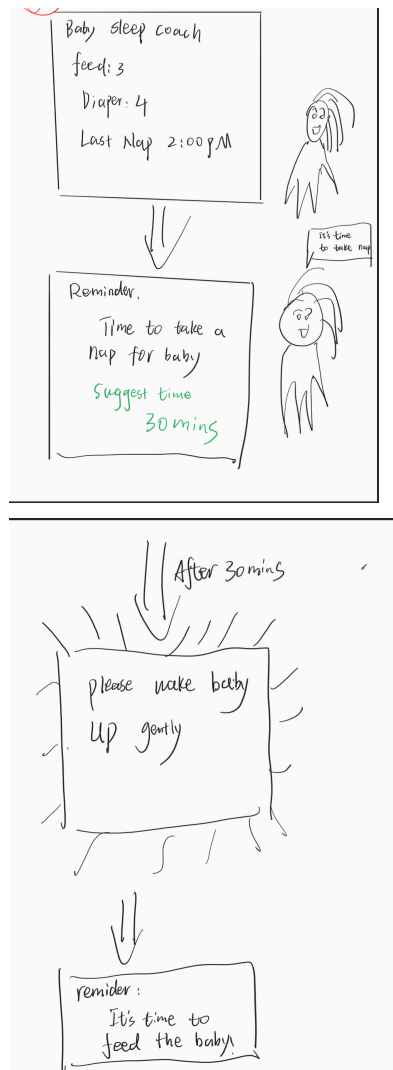
User Group: Senior Citizens (60+)

9. An AI reminder system could help seniors with early-onset dementia manage daily routines like medication and appointments.
10. A phone coach powered by AI could guide older adults step-by-step on how to use smartphones and digital tools.
11. An AI-powered social companion could reduce isolation and support decision-making by suggesting ideas or initiating video calls with friends or volunteers who can support with advice and tasks (inspired from *Be My Eyes* for visually impaired).

12. A financial planning assistant could help seniors organize wills, healthcare forms, and financial records securely.
13. A caregiving support app could explain medical instructions and detect medication errors using computer vision.
14. An AI reading assistant could read text out loud for elderly people with vision disabilities.

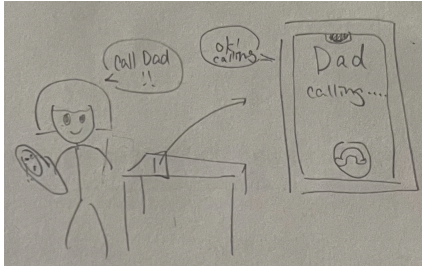
User Group: New Parents

15. An AI-powered baby monitor could detect unusual crying patterns and suggest possible causes, such as hunger or discomfort.
16. A sleep coach could analyze a baby's sleep data and provide gentle bedtime routine suggestions for parents.



17. A nutrition assistant could generate baby-friendly meal plans as children grow, tailored to health needs.
18. A virtual parenting coach could answer real-time questions about developmental milestones.
19. A financial planning tool could help new parents predict costs of childcare, diapers, and family needs.
20. An information hub could provide detailed instructions for new parents on newborn care tasks.

21. A hands-free baby care assistant could let parents give voice commands to set reminders or make calls while holding the baby.

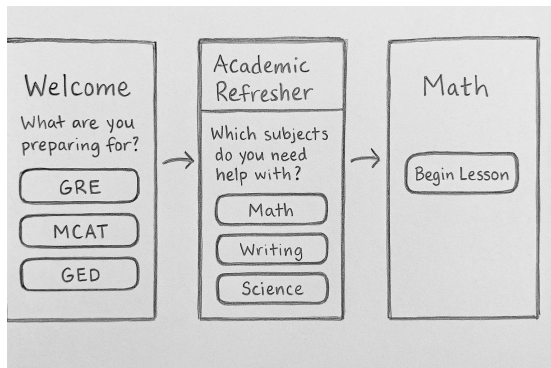


User Group: Empty Nester Parents

22. An AI lifestyle assistant could suggest new hobbies and activities for parents adjusting to more free time.
23. A loneliness companion app could detect mood and recommend uplifting social events or connections.
24. A travel-planning assistant could generate itineraries for newly retired parents looking to explore.
25. A memory-capturing tool could organize and share family photos into digital storybooks.
26. A personalized fitness motivator could generate health routines to keep parents active.

User Group: Adults Going to College for the First Time

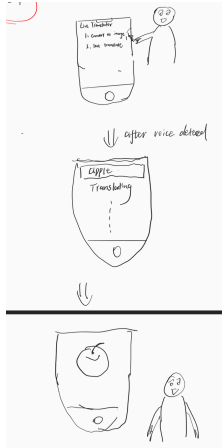
27. An academic refresher app could fill gaps in math, writing, and science knowledge for adult learners looking to return to school.



28. A study planner could adapt to busy schedules by balancing coursework with work and family responsibilities.
29. A writing coach could provide real-time grammar corrections and clarity suggestions on essays.
30. A financial aid assistant could explain complex forms and generate personalized checklists.
31. A peer-connection tool could match adult learners with similar backgrounds for study support.
32. A time management assistant could predict the best study times by analyzing work and family schedules.
33. A community connector could match adult learners with others in similar situations for emotional support.

User Group: ESL Learners

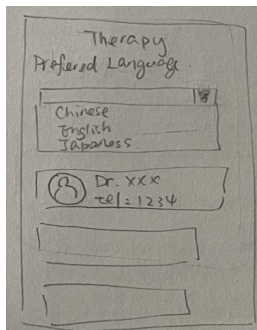
34. A live translation app could combine images and definitions with spoken translation for better comprehension.



35. A pronunciation coach could analyze recorded speech and provide personalized corrections.
36. A reading buddy could simplify complex text and explain vocabulary through AI-generated examples.
37. A conversation simulator could role-play everyday scenarios like grocery shopping or doctor visits.
38. A professional writing helper could suggest clearer phrasing for workplace communication.

User Group: People Struggling with Mental Health

39. An AI journaling app could detect negative thought patterns and suggest healthier reframes.
40. A virtual companion could practice cognitive behavioral therapy (CBT) exercises interactively.
41. A mood tracker could analyze voice tone and recommend stress-relieving activities.
42. An AI meditation guide could generate personalized relaxation scripts based on user stress levels.
43. A social connection tool could recommend low-pressure activities to help reduce loneliness.
44. A therapy-matching assistant could connect users with therapists who speak their preferred language.



User Group: DMV Employees

45. An AI queue manager could predict wait times and balance staff workloads to reduce stress.
46. A document checker could scan and flag incomplete forms before customers reach the counter.
47. A voice-to-text assistant could transcribe and organize customer interactions for better record-keeping.
48. A training simulator could let employees practice handling frustrated customers using AI role-play.
49. A scheduling optimizer could recommend staggered break times to reduce burnout.

User Group: Students with Learning Disabilities

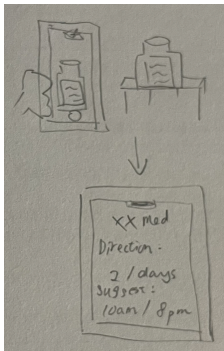
50. An AI reading assistant could simplify text for dyslexic students while preserving meaning.



51. A writing support tool could convert speech to text with automatic grammar adjustments.
52. A math helper could visualize problems for students with processing challenges.
53. A classroom companion could summarize lecture notes into simpler formats.
54. A task planner could break down assignments into smaller steps and send reminders.

User Group: Caregivers for Elderly Relatives

55. A pill management app could scan prescription bottles and create personalized reminders to prevent missed or double doses.



56. An AI health tracker could analyze wearable data such as heart rate and sleep to alert caregivers if something looks unusual.
57. A speech-to-text and summarization tool could record doctor visits and turn complex instructions into simple checklists.
58. A family planner could use AI scheduling to coordinate multiple relatives sharing caregiving responsibilities.

Discussion of Scoring (Specificity, Beneficence, Risk, AI Performance)

In our scoring discussion, we began by narrowing our 50+ brainstormed ideas down to ten main ones, which we did through allowing each of us to independently write down our favorite five (and then referencing which ones had any overlap). The discussion then included each member having a few minutes of un-interrupted time to advocate for or explain the reasoning behind the personal favorites and what a potential prototype would look like for a certain idea, allowing us to narrow it down to an AI-powered social companion for the elderly, a live-translation app for ESL learners, and an AI reminder system that caregivers could use when taking care of seniors with dementia.

The discussion for scoring criteria allowed us to evaluate that the live-translation app for ESL learners also had high benefit and clear specificity, but it felt unrealistic for our timeframe since it would require large datasets and advanced training to accurately convert text and product images depending on the languages converted. The

AI reminder system for seniors with dementia addresses an important need and was technically feasible, but it introduced major risks if errors occurred with medication reminders. The AI-powered social companion for the elderly scored well because it targeted a clear user group (elderly seniors that have access to a device and might be struggling with isolation), has the potential to drive social benefit through providing connection and mental well-being, and while it can carry privacy risk, the goal is to prototype a feature to verify volunteers. The scoring exercise helped us see that the social companion app offered a better balance of specificity, beneficence, low risk, and realistic AI performance, as well as is of high interest to the team members since it can be incredibly useful for senior citizens if executed correctly.

Idea Selection

The idea that the group has chosen to work on is an AI-powered social companion for elderly people, especially those who might be struggling with loneliness or a lack of support. It is meaningful because it directly addresses the growing problem of senior loneliness and social disconnection. We also drew inspiration from the app called *Be My Eyes*, which is designed for volunteers to sign up and answer calls from those with visual impairments or disabilities. Instead of solely relying on live people to provide support, the project would also integrate artificial intelligence to offer more consistent support.

Project Description for Selected Idea

- **Target User Group**

The target user group for the project is senior citizens (aged 60+) who do have access to the internet and are able to somewhat navigate it sufficiently. In addition, the group more so includes those who live alone, are separated from families, or might need more support with mundane (and some higher level) tasks and questions. The aging population is becoming more and more relevant as it is continuing to increase, and immigration restrictions and healthcare expenses are making it so that there are less caregivers to support the elderly. The hypothetical needs and wants stem from some older people having limited mobility and smaller social circles, making it difficult for them to stay socially active and connected. For those who might want to improve the quality of life, strengthen social connections, and receive support with decisions or navigating different platforms, . They may also struggle with using smartphones to communicate with family members or face challenges in managing calendars and schedules. These are the problems we aim to solve.

- **Problem Description & Context**

The problem is that a lot of elderly people and senior citizens are struggling with loneliness and feeling disconnected from society due to living alone, having health or mobility issues, or being distant from (or having lost) family and friends, which can lead to memory decline and difficulty making decisions on a day-to-day basis. The challenges often occur in the home environment, or even in places like grocery stores (where even young people can struggle with decision fatigue), where seniors may spend long hours with limited social interaction, and are compounded by barriers such as limited transportation, lack of nearby community programs, or difficulty navigating digital tools. There are also factors such as lower familiarity with technology, reduced energy or mobility, and varying levels of education and digital literacy influence how seniors engage with existing supports. There are some services that do already exist, such as phone hotlines, community centers, and volunteer-based platforms like *Be My Eyes* (although that is directed toward the visually disabled rather than elderly), many require either high digital skills or the availability of live human volunteers, which are not always accessible. The project could highlight the need to explore ways of making consistent social connection and decision support more approachable and sustainable for this user group.

- **How does your application address the theme of intelligent user interfaces? (one to two sentences)**

The application addresses the theme of intelligent user interfaces by integrating AI to support the elderly through providing suggestions on simple questions or tasks, teaching how to navigate different tasks (for example, wanting to set up a TV remote), and connecting them with family members or volunteers.

GenAI Reflection

The team did use Generative AI (in specific, OpenAI's ChatGPT), to support turning the 50+ brainstormed ideas into full sentences. In class and in a meeting afterward, we had already developed the potential target user groups and project ideas within each theme and category, but were able to share that with the LLM to save time on the repetitive task of turning each bullet point into a full sentence. There were also some limitations, such as that even without us necessarily asking for it, ChatGPT tended to make suggestions that sometimes leaned too optimistic or technologically advanced for our project timeframe (e.g., implying access to very large datasets or computer vision models). It was important for us to review through the output to make sure that it was adjusted or scaled back to fit the assignment requirements and not get ahead of itself for the project.